

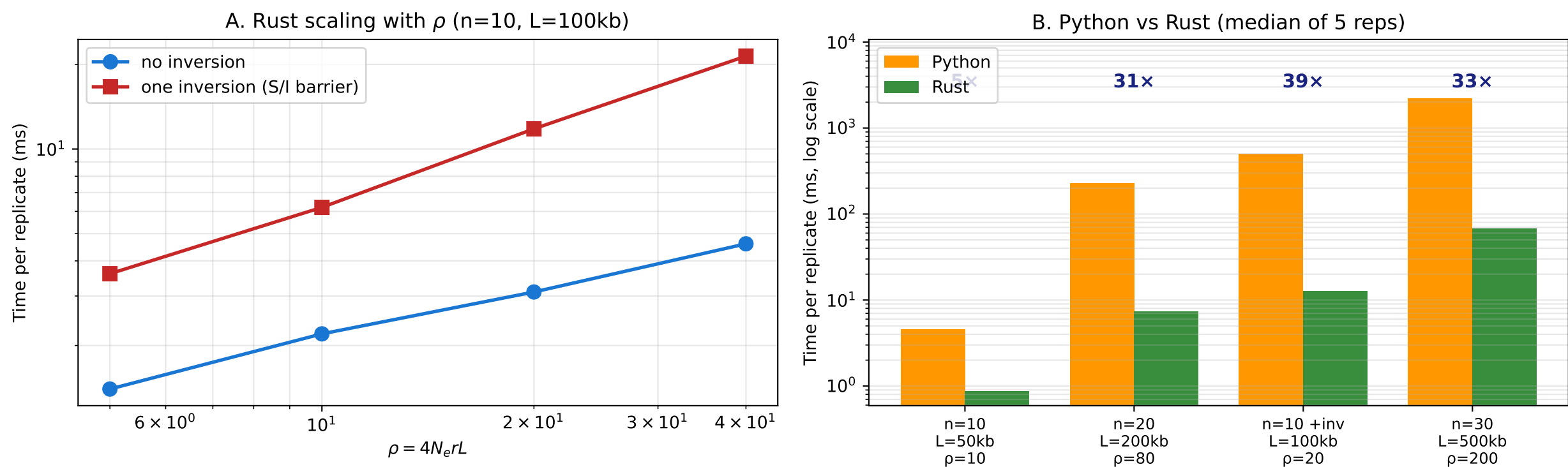


Figure 7. *msinv* (Rust) performance. (A) Wall-clock time per replicate scales sublinearly with recombination $\rho = 4N_e rL$. The S/I class barrier adds modest overhead from per-segment class tracking and gene-flux events. (B) Python vs Rust on four representative scenarios; speedup (above bars) grows with problem size. The Rust core is the default backend in *msinv* $\geq 0.4.0$; the legacy Python implementation is retained for cross-validation. Panel B: median of 5 replicates per scenario.